

Module 4 — Assignment 01

Movie Collection API

Topics: *GET* • *POST* • *Path Parameters* • *Query Parameters*

Total Marks: 100

Overview

You will build a Movie Collection API from scratch using FastAPI. No database required. Load initial data from a `movies.json` file.

Data Model

Each movie has the following fields:

Field	Type	Description
<code>movie_id</code>	<code>int</code>	Sent by the client in the POST request body
<code>title</code>	<code>str</code>	Title of the movie
<code>director</code>	<code>str</code>	Director's name
<code>genre</code>	<code>str</code>	One of: "action", "comedy", "drama", "thriller"
<code>duration_minutes</code>	<code>int</code>	Total runtime of the movie in minutes
<code>rating</code>	<code>float</code>	Movie rating (e.g. 4.5)

`movies.json` File

You must create a file named `movies.json` in your project folder. When the app starts, read this file and load the movies into your in-memory list. This means your GET endpoints work immediately without needing to POST first.

`movies.json` content:

```
[
  {
    "movie_id": 1,
    "title": "Inception",
    "director": "Christopher Nolan",
    "genre": "thriller",
    "duration_minutes": 148,
    "rating": 4.8
  },
  {
    "movie_id": 2,
    "title": "The Hangover",
    "director": "Todd Phillips",
    "genre": "comedy",
    "duration_minutes": 100,
    "rating": 4.2
  },
  {
    "movie_id": 3,
    "title": "The Pursuit of Happyness",
    "director": "Gabriele Muccino",
    "genre": "drama",
    "duration_minutes": 117,
    "rating": 4.7
  }
]
```

How to load it in main.py:

```
import json

with open('movies.json', 'r') as f:
    movies = json.load(f)
```

Required Endpoints

1. GET /movies

GET	/movies	20 Marks
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Return all movies currently in the movies.json file.

2. GET /movies/{movie_id}

GET	/movies/{movie_id}	25 Marks
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Return a single movie matching the given movie_id.

- If found: return the movie object with status 200.
- If not found: return a meaningful error message with status 404. Do not crash.

3. POST /create_movies

POST	/create_movies	30 Marks
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Add a new movie to the collection.

- The client sends all fields in the request body, including movie_id.
- Append the new movie to the in-memory list.

- Return the newly created movie object with status 201.

4. GET /movies/sort

GET	/movies/sort	25 Marks
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Return all movies sorted by a field, in a given direction.

Query Parameters:

Parameter	Allowed Values	Default
sort_by	duration_minutes, rating	rating
order	asc, desc	desc

Example: /sort?sort_by=duration_minutes&order=asc

Marking Summary

Task	Endpoint	Marks
1	GET /movies	20
2	GET /movies/{movie_id}	25
3	POST /create_movies	30
4	GET /movies/sort	25
Total		100